

Question	Marking Guidance	Mark	Comments
08.1	1. Add 1 part (bacteria) culture to 9 parts (sterile) liquid (to make 10^{-1} dilution); 2. Mix (well); 3. Repeat using 9 parts fresh (sterile) liquid and 1 part of 10^{-1} and 10^{-2} dilutions to make 10^{-3} dilution; OR Add 1 part 10^{-1} (suspension) to 99 parts (sterile) liquid (to make 10^{-3} dilution);	3	1. and 3. Accept water / nutrient / broth for liquid 2. Accept stir 3. Reject 1 part (undiluted) culture added to 999 parts liquid
08.2	$3.75 \times 10^9 / 3\,750\,000\,000$;	2	Accept for 1 mark: $3750\,000 / 3.75 \times 10^6$ (cells per mm^3) OR 3.75×10^{12} (wrong volume conversion) OR 3750 (cells per mm^3 of diluted culture) OR Evidence of using correct dilution conversion and correct volume conversion, i.e., $\times 1000$ and $\times 1000$
08.3	1. Count unlikely to be accurate / repeatable / reproducible / reliable; 2. Because too many cells; OR Because cells overlapping / not spread out;	2	

Question	Marking Guidance	Mark	Comments
08.4	1. Tetracycline used more often / in higher doses; 2. Resistant bacteria more likely to (survive and reproduce and) pass on allele/gene for (tetracycline) resistance; OR 3. More / higher frequency of mutations (for tetracycline resistance); 4. (so) gene passed on to more bacteria; OR 5. Tetracycline used over longer time period; 6. More time for (chance) mutation to occur / for selection to occur;	2	Ignore reference to resistant animals Ignore reference to immunity 3. Reject reference to mutation being caused by use of antibiotic
08.5	No selection against resistant bacteria / resistance gene/allele; OR Bacteria pass on (resistance) gene / allele when they reproduce; OR Bacteria resistant to tetracycline are passed on from one generation of farm animals to the next (probably via faeces); OR Environment does not change, so stabilising selection occurs;	1	Accept no selection to get rid of it Reject reference to mitosis or immunity